

Streamlining Amazon Seller Operations Through Advanced Automation and AI (Amauton)

Project Team

Ali Khizar	21I-0514
Irtiq Haider	21I-0733
Talha Shahzad	21I-0540

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Supervised by

Dr. Hassan Raza

Co-Supervised by

Mr. Ahmed Raza



Department of Computer Science

**National University of Computer and Emerging Sciences
Islamabad, Pakistan**

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Chapter 1

Introduction

More smart instruments are needed by sellers to be ahead in the fast moving world of e-commerce. The aim of this project is to create an automation system for automating vital activities for Amazon Seller Accounts focusing on UK market. The system will help sellers in managing their accounts more easily and to increase profitability by features like dynamic pricing, sales analytics, and product discovery. The system will alert users to important changes, moreover it will analyze trends in a single product category and comparing products from top five UK retailers, including Tesco and Sainsbury's. The system will save the seller's time in the market, reducing hard tasks, and helping them to make better choices, thus increasing their efficiency and competitiveness in the long run. The main objectives, purpose, tools and technologies of the project are all outlined in this document.

1.1 Problem Statement

Managing and staying competitive in this dynamic and ever-developing business world is becoming harder day by day. Amazon is ranked as the number one E-commerce and Shopping website[?], Amazon Seller Account is one of the many businesses associated with it. Seller operations involve a lot of manual labor and are taking time. Amazon is continuously looked through by sellers for beneficial products, comparing pricing with different retailers and calculating profit margins.

Ineffectiveness, lower profitability, and lost opportunities are caused by these labor-intensive operations. Sellers find it challenging to maintain their competitiveness in a quickly changing market due to lack of automation and rapid judgements. Sellers are unable to quickly adapt their strategies to stay competitive because of this, making it harder for businesses to find profitable products and set up competitive prices with the lack of real-time analytics and automation systems

1.2 Scope

This project's scope is centered on the UK market and includes developing a system that streamlines important tasks for Amazon Seller Accounts. With the intent of streamlining seller operations, the system will have features like dynamic pricing, sales analytics, and product discovery.

The product discovery feature will provide niche-specific recommendations, assisting sellers to identify profitable products across various categories. The solution will analyze products from the top five supermarkets, including Tesco and Sainsbury's, for more comprehensive market research so that sellers remain competing and make well-informed decisions.

Real-time price adjustments will be made automatically by dynamic pricing in response to competition activity and sales analytics will give insightful data on profitability, sales patterns, and product performance. The solution will first monitor trends in a single product category and alert sellers to significant modifications and updates in the market. This feature can be extended to include additional categories in the future.

Supplier sourcing and acquisition will happen more quickly, saving sellers time and energy so they can focus on expanding their businesses. The system will also automate routine tasks like listing updates and pricing modifications, assisting vendors in maintaining their competitiveness. For Amazon sellers, the idea is to maximize revenue, enhance decision-making, and save time by automating some of their most time-consuming operations.

1.3 Modules

1.3.1 Module 1: Product Discovery and Extraction

This module helps in finding and collecting relevant product data using different search options.

1. Search by keywords
2. Apply filters set by the user
3. Extract data from product listings
4. Analyze market trends to find popular products

1.3.2 Module 2: Supplier Sourcing

This module focuses on finding suppliers and making sure products match the listings.

1. Identify suitable suppliers
2. Use AI to check if the product images and titles are consistent across different sources

1.3.3 Module 3: Profitability Analysis and Listings

This module calculates profit margins and helps create product listings.

1. Calculate profit margins
2. Create product listings automatically, including:
 - Generate product titles
 - Write descriptions and bullet points
 - Set pricing

1.3.4 Module 4: Dynamic Pricing and Competitor Analysis

This module dynamically adjusts prices and monitors competitors in real-time.

1. Dynamic pricing engine
2. Real-time competitor analysis

1.3.5 Module 5: Sales Analytics

This module provides insights and visual tools to analyze sales performance.

1. Data visualization
2. Generate actionable insights
3. Sales analytics dashboard

1.4 User Classes and Characteristics

Identify the various user classes that you anticipate will use this product, and describe their pertinent characteristics.

Table 1.1: User Classes and Characteristics Table

User class	Description
Amazon Sellers	Amazon sellers are the one responsible for managing product listings, pricing, and supplier relationships. Identifying the best products, listing products, and keeping an eye on market trends are some of the key activities they manage on a daily basis through their interactions with the system. The technology helps sellers find profitable items and maintain their competitive edge in the market. Sellers can focus on strategic growth by frequently using the platform for real-time pricing adjustments, sales analytics, and market alerts, while the system takes care of operational activities that usually take a lot of time and effort.
Data Analysts	Data Analysts are responsible for analyzing sales data, profitability patterns, and market dynamics. To perform in-depth analysis and demonstrate big datasets, they need sophisticated tools. Analysts may provide in-depth insights into competitive environments and niche markets by utilizing the system's comprehensive sales analytics, trend tracking, and data export features. With the use of these insights, sellers are better equipped to make profitable and growth-oriented strategic decisions.
Retailers and Suppliers	Amazon sellers can purchase goods from retailers and suppliers, and they may also provide an interface to the system that allows sellers to view thorough stores information. This comprises information about the product name, description, cost, and delivery location. Retailers and suppliers offer the system ability to determine product pricing and availability by supplying the required information, which helps Amazon sellers make informed choices and effectively manage their product sourcing and hunting.

Example: User Classes and Characteristics

Chapter 2

Project Requirements

This chapter describes the functional and non-functional requirements of the project.

2.1 Use-case/Event Response Table/Storyboarding

The selection of the requirement gathering technique we will be using is Event Response table, due to the reason most of our task are performed at back-end.

Table 2.1: Event Response Table

Event	System State	Response
User enters the URL and opens the application	System have not started yet	System will fetch required information for DB and displays user Dashboard.
User navigate to Seller Operation option.	User dashboard is displaying with other Menu and features.	• System will find the trending products and display them with relevant information. • System will display an input field for the user to enter product description.
User enters the product detail in the input field and specify the filters to be applied	Market trends with some product information is displayed.	System will narrow down the input based on filters and start searching for the product
Sourcing of different suppliers based on input product to fetch necessary information.	System has taken the input and applied the user custom filters.	• System will do profit comparison, it will stop working if the product is not profitable and vice versa. • System will fetch the information from different specified stores for further processing.
Comparing the fetched products with the product listed on amazon.	System have fetched the information from different stores containing title, description, price and images.	System will match the text and images by using comprehensive methods and techniques.
Presenting of input product with their relevant suppliers and fetched information.	System completed the comparison and found the suitable matches.	System will display the product with their profitable suppliers along with necessary information in an interactive way and asks for Listing options
User selection of product listing on amazon store.	All required information is being presented and system asks for listing options.	• System will perform automatic listing of product on store by adding the required information in the fields. • System will redirect the user to amazon site to manually fill the required information.
Alterations in the prices of amazon competitors	Product are listed on the user amazon store.	System utilizing dynamic pricing engine will set price to be in competitive race otherwise drop the product.

2.2 Functional Requirements

This section describes the functional requirements of the system expressed in the natural language style.

2.2.1 Product Discovery and Extraction

Following are the requirements for Product Discovery and Extraction:

1. System shall display the user current market trends in human readable form with necessary information.
2. System shall allow user to enter product details and should cater any style of input.
3. System shall allow user to apply custom filters based on the option provided.
4. System shall correctly apply the filters and move from generalization to specialization for efficient sourcing.
5. System shall search for the entered product on different stores and correctly retrieve the required information.

2.2.2 Supplier Sourcing

Following are the requirements for Supplier Sourcing:

1. System shall find the entered product on different stores specified in the scope.
2. System shall identify the suitable supplier by refining based on the profit margins.
3. System shall retrieve the required information about the product from the selected suppliers.
4. System shall efficiently compare and match the relevant information such as title, price and images.

2.2.3 Profitability Analysis and Listings

Following are the requirements for Profitability Analysis and Listings:

1. System shall calculate profit margins leveraging different sales and analytics techniques.

2. System shall automatically manage the listing of product on user's amazon store.
3. System shall generate the required information about the product and efficiently add the title, description, price and image in the right place.

2.2.4 Dynamic Pricing and Competitor Analysis

Following are the requirements for Dynamic Pricing and Competitor Analysis:

1. System shall relentlessly looks for changing competitors prices.
2. System shall constantly adjust the prices to be in the competitive race.
3. The system shall drop the product if it lacks profitability or sales activity.

2.2.5 Sales Analytics

Following are the requirements for v:

1. System shall present the sales and profit of different products in interactive manner.
2. System shall generate effective insights from the user sales history.
3. The system shall provide different visualization for easy and better understanding.

2.3 Non-Functional Requirements

This section specifies nonfunctional requirements.

2.3.1 Reliability

1. In case of failure in any kind of operation such as fetching data from DB, listing product and product hunting cycle, System must rollback and try 2-3 times before logging the error.
2. In case of continuous failure system must notify the user or administrator through email.
3. The system shall have a minimum MTBF of 30 days to ensure continuous operations without unexpected downtime.

2.3.2 Usability

1. System should be user friendly and interactive, feature should be easily accessible and understandable.
2. System shall have a logical flow and each feature must perform what it is entitled to.
3. The system shall record repetitive tasks to increase the efficiency of interaction

2.3.3 Performance

1. Website must be within 10 seconds along with the text and graphics.
2. System shall perform specific task such as pricing adjustments and product matching in logical time frame such as within 2 minutes.

2.3.4 Security

1. Website must be secure from different attacks which might result in data leak.
2. It must use strong and secure methods to store passwords in DB.
3. User information must be handled efficiently, in case of data breach, with no loss tolerance.
4. It must support multi-factor authentication and email confirmations to manage access.

2.4 Domain Model

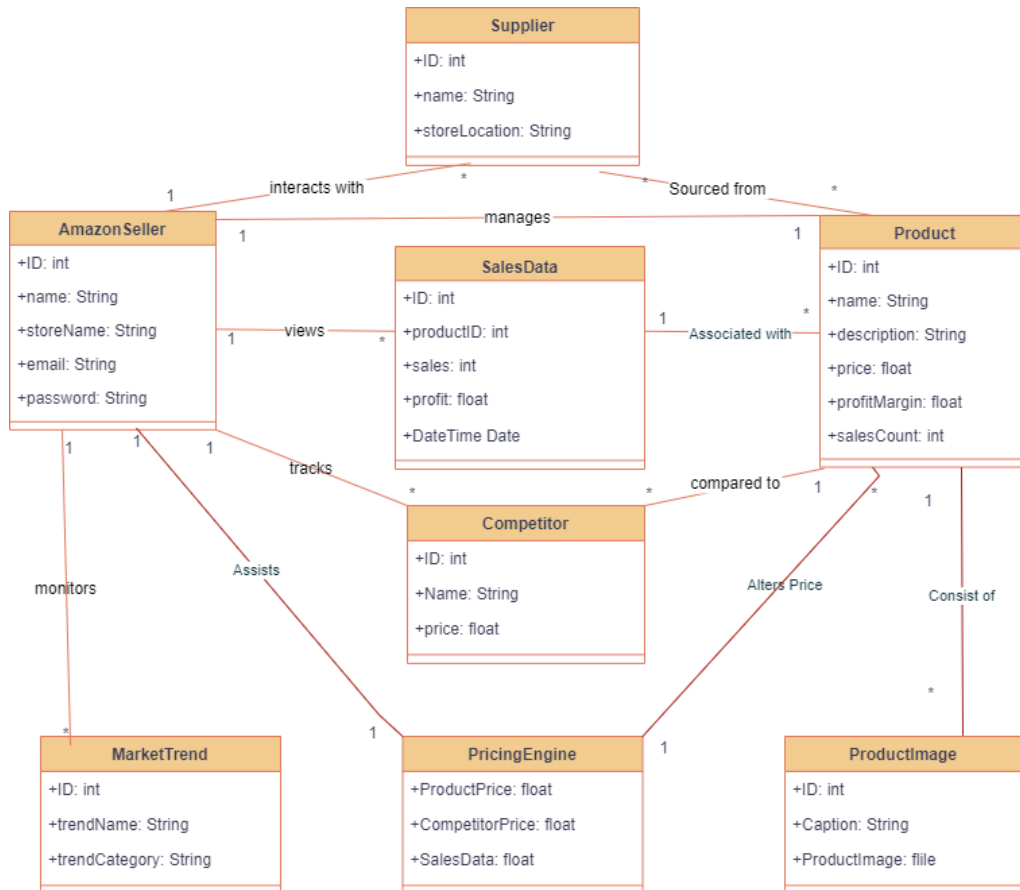


Figure 2.1: Domain Model of Amauton

Chapter 3

System Overview

This Chapter provides a general description of the functionality, context, and design of your project.

3.1 Architectural Design

The architecture pattern we will be using is Micro-service architecture. Starting from above is the DB, then Micro services, API gateway for communication between them and in the end UI for users.

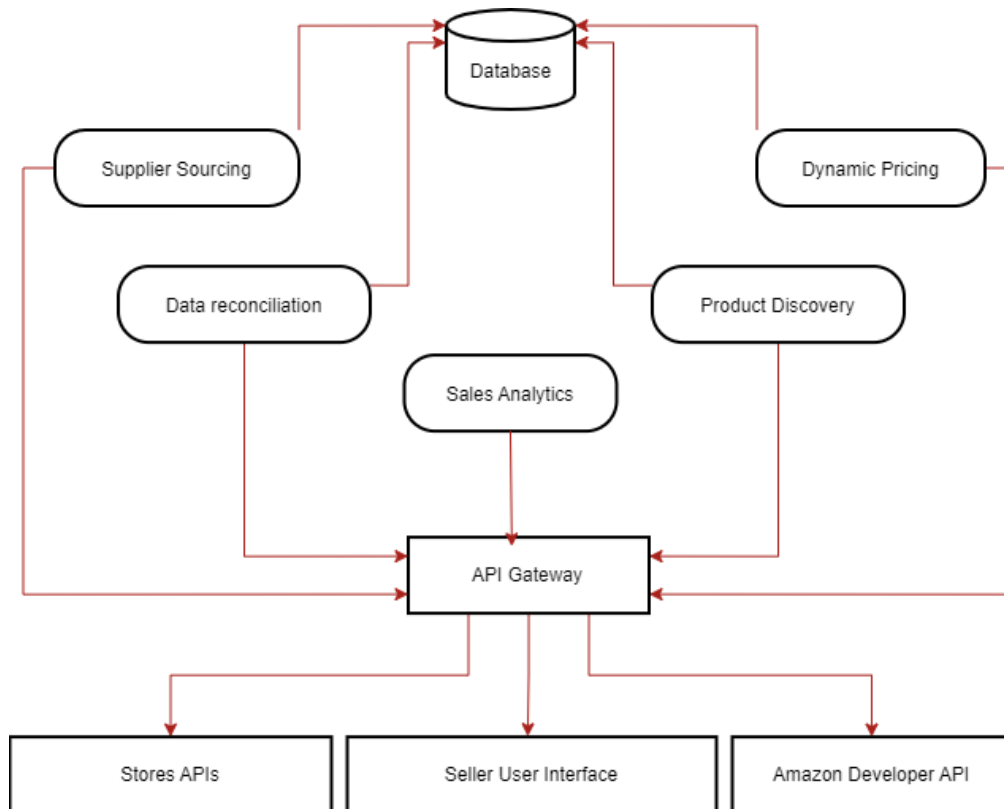


Figure 3.1: Architecture pattern

3.2 Design Models

Design Models for Procedural Approach

The applicable models for the project using procedural approach.

3.2.1 Activity Diagram

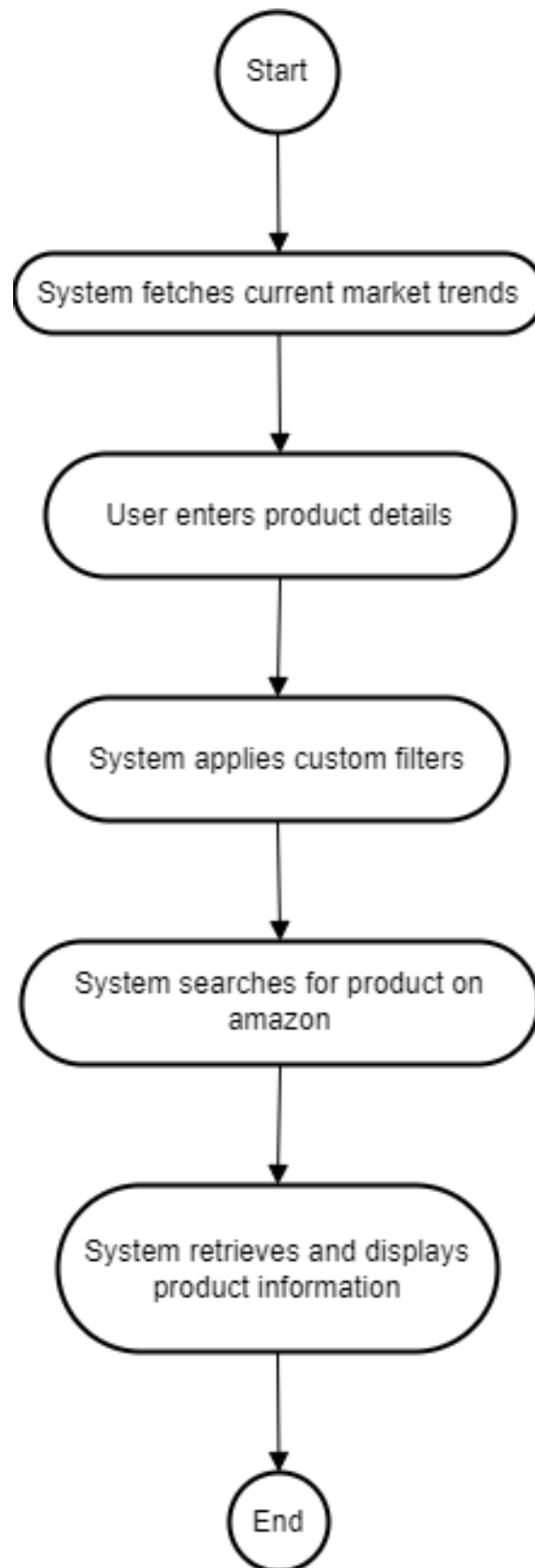


Figure 3.2: Product Discovery and Extraction Module

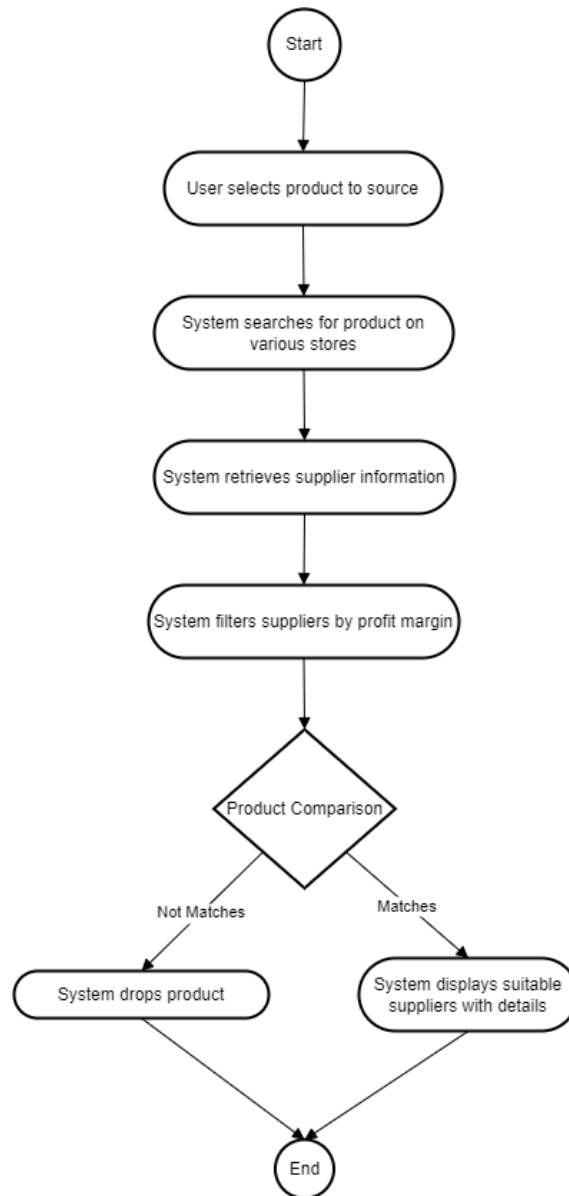


Figure 3.3: Supplier Sourcing Module

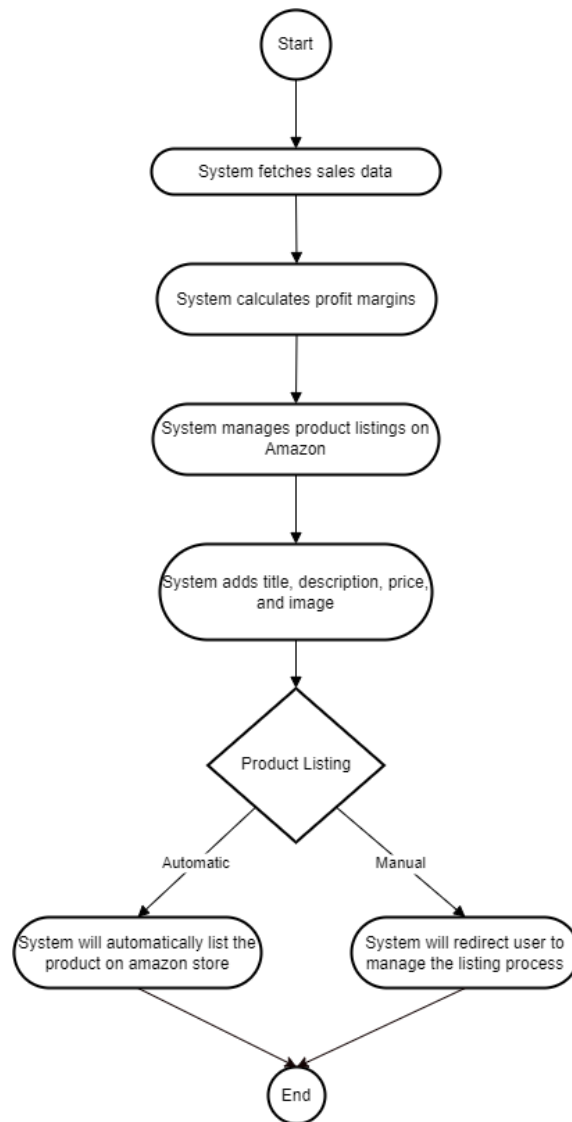


Figure 3.4: Profitability Analysis and Listings Module

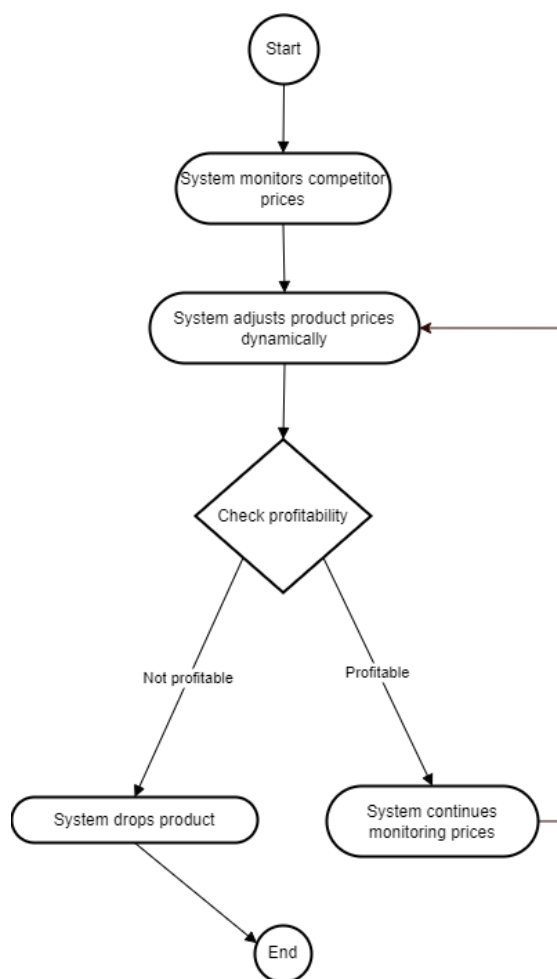


Figure 3.5: Dynamic Pricing and Competitor Analysis Module

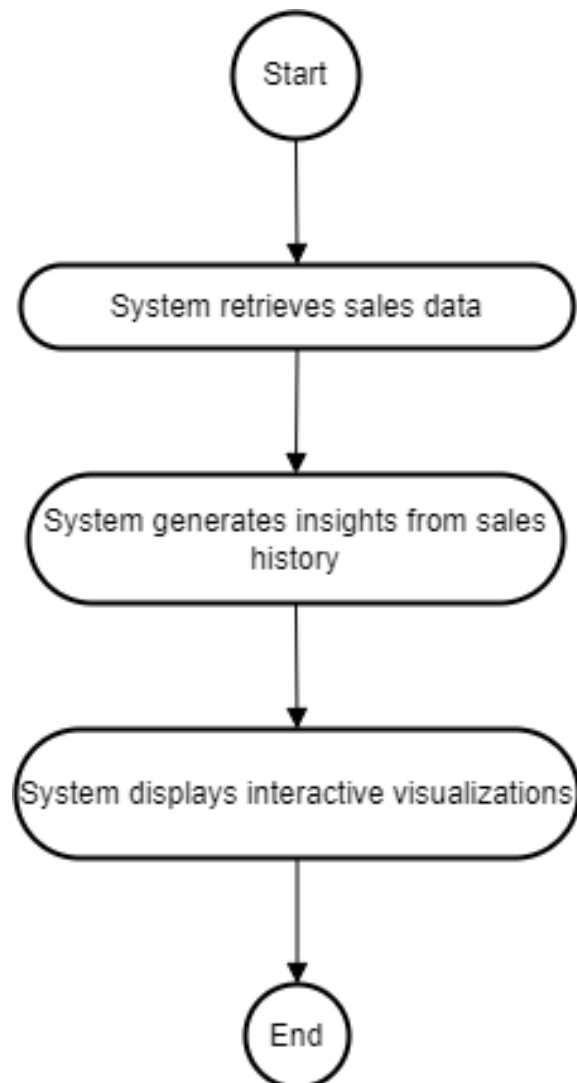


Figure 3.6: Sales Analytics Module

3.2.2 Data Flow Diagram

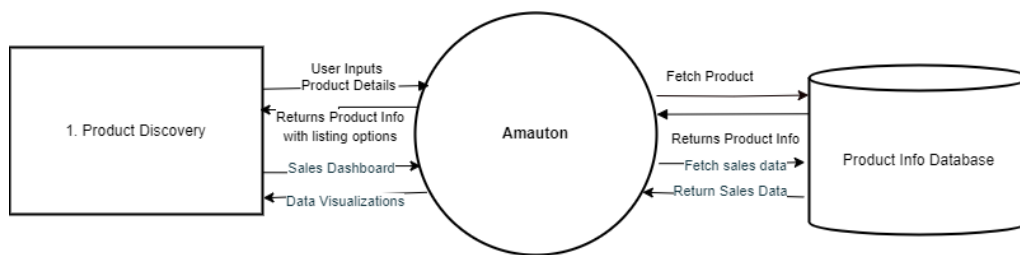


Figure 3.7: Data Flow Diagram Level 0

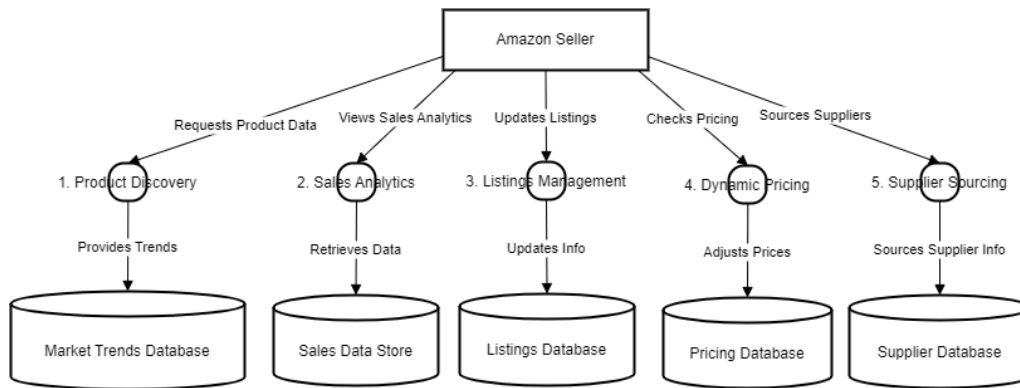


Figure 3.8: Data Flow Diagram Level 1

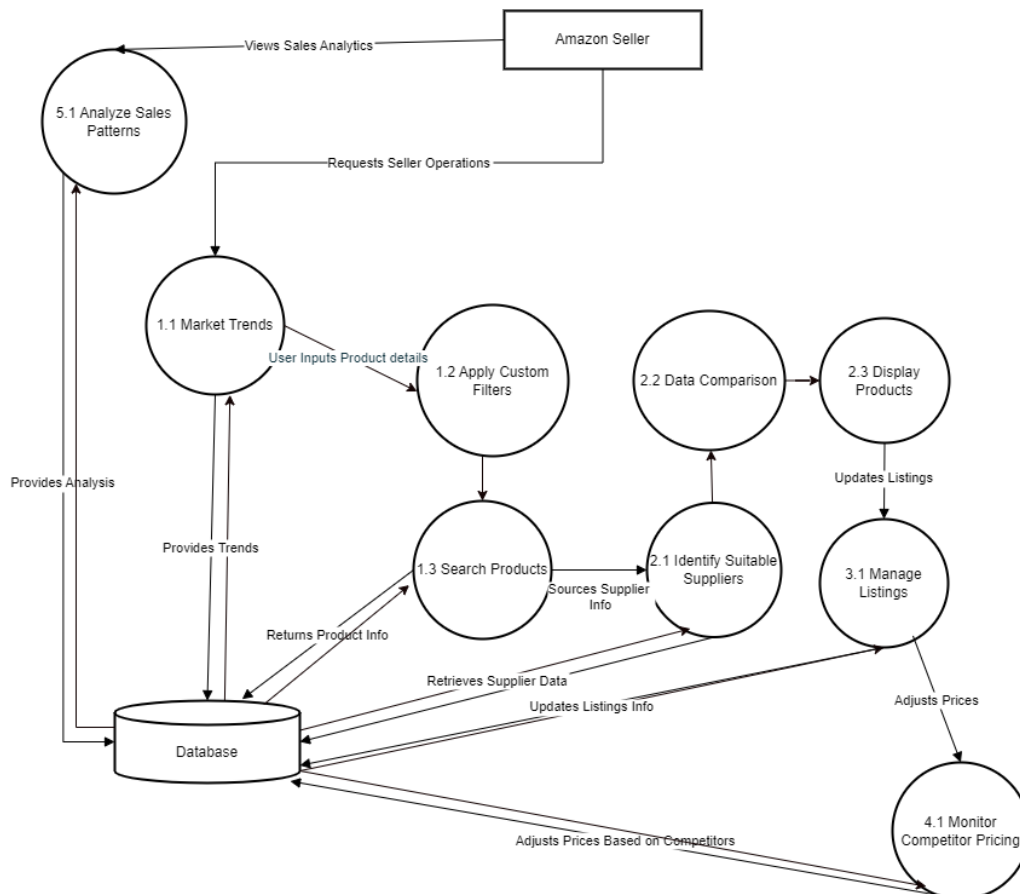


Figure 3.9: Data Flow Diagram Level 2

3.2.3 System-level Sequence Diagram

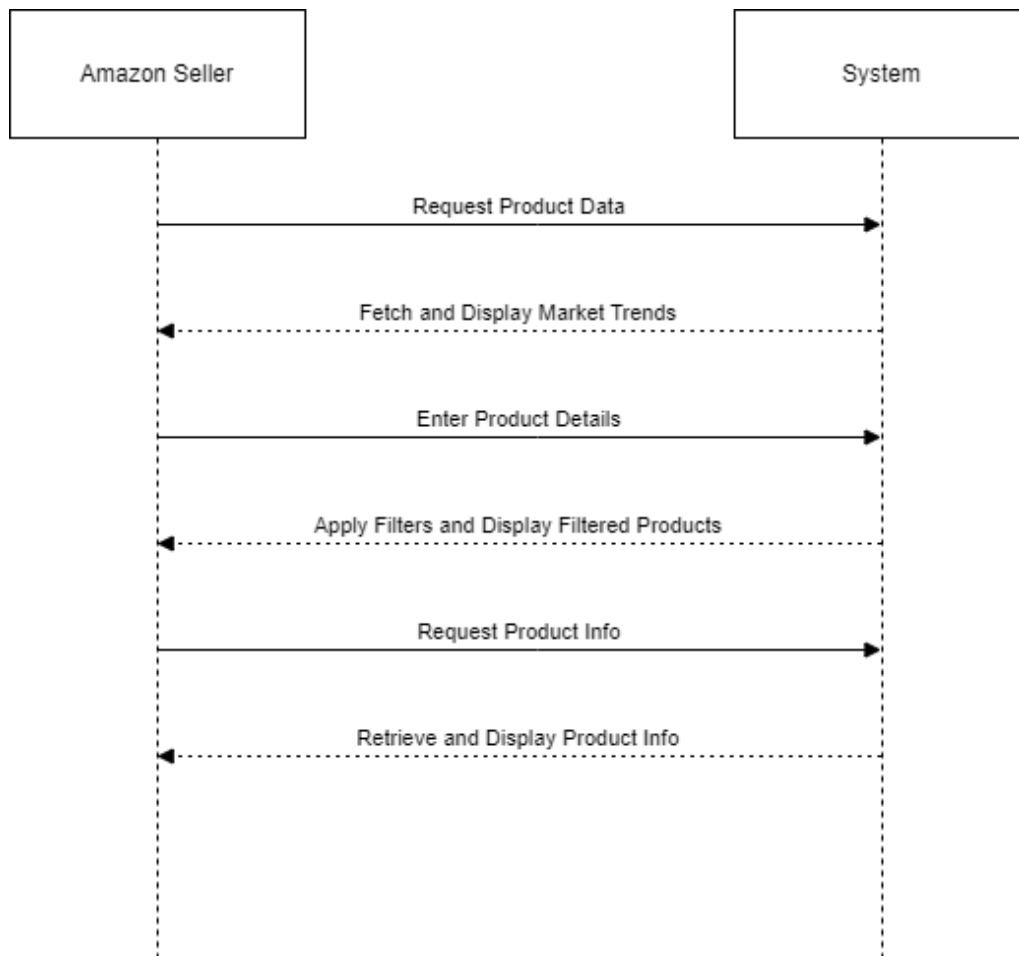


Figure 3.10: System sequence diagram 1

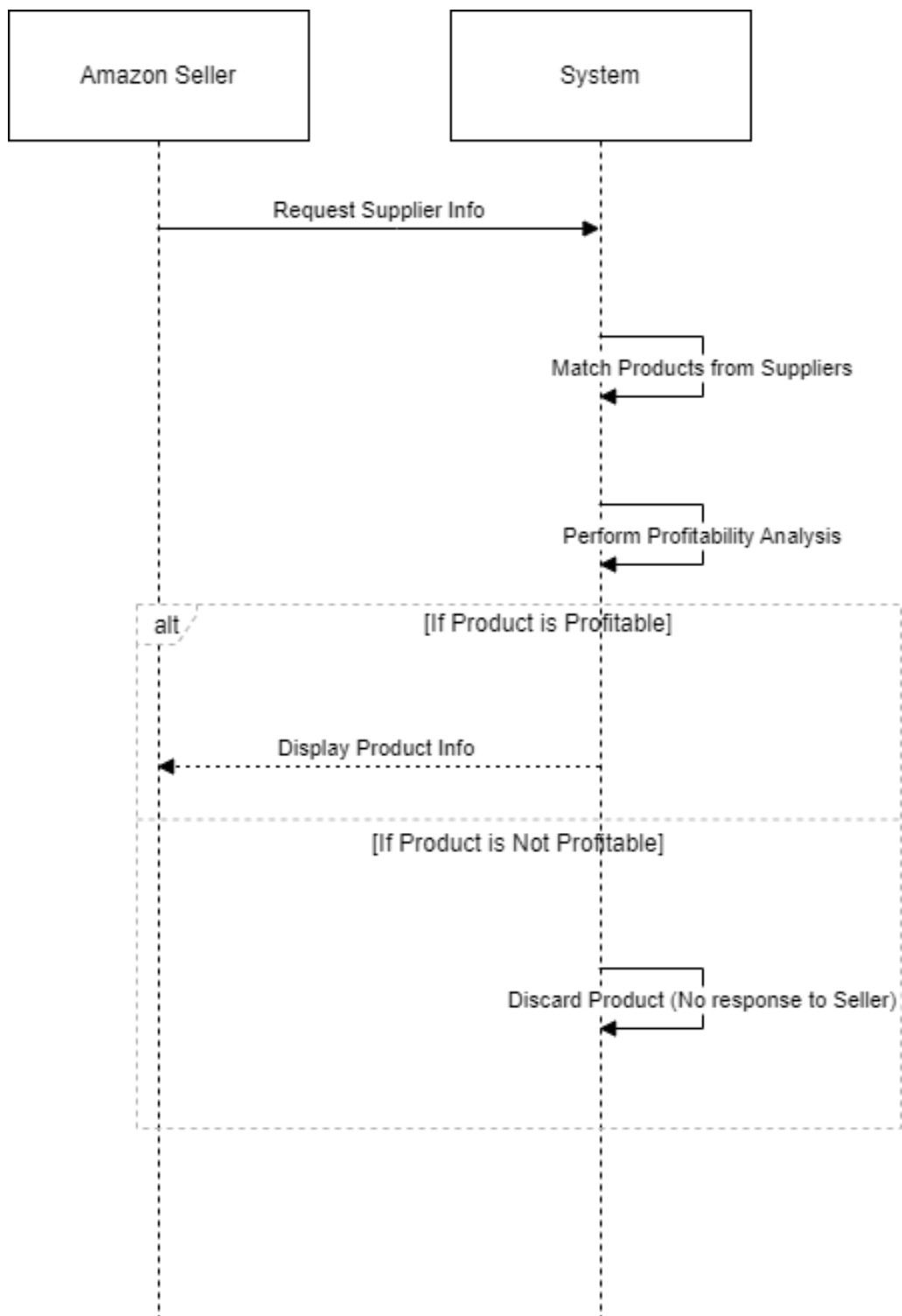


Figure 3.11: System sequence diagram 2

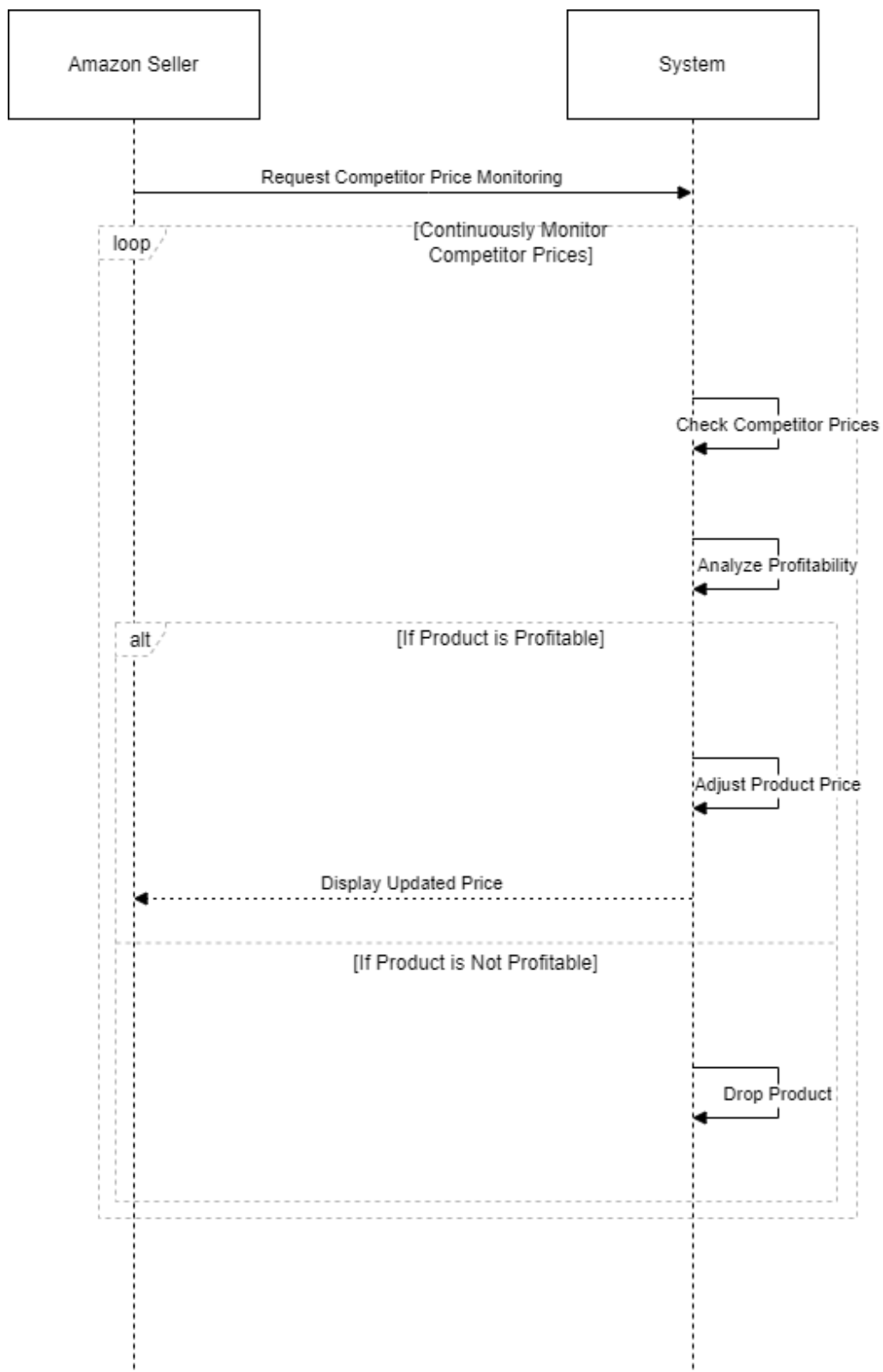


Figure 3.12: System sequence diagram 3

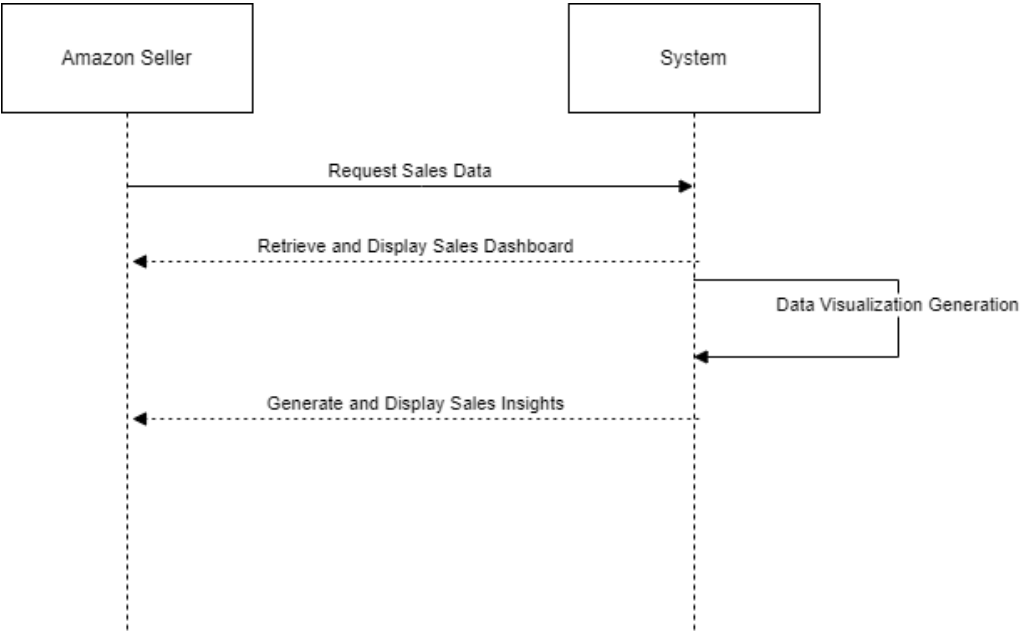


Figure 3.13: System sequence diagram 4

3.2.4 State Transition Diagram

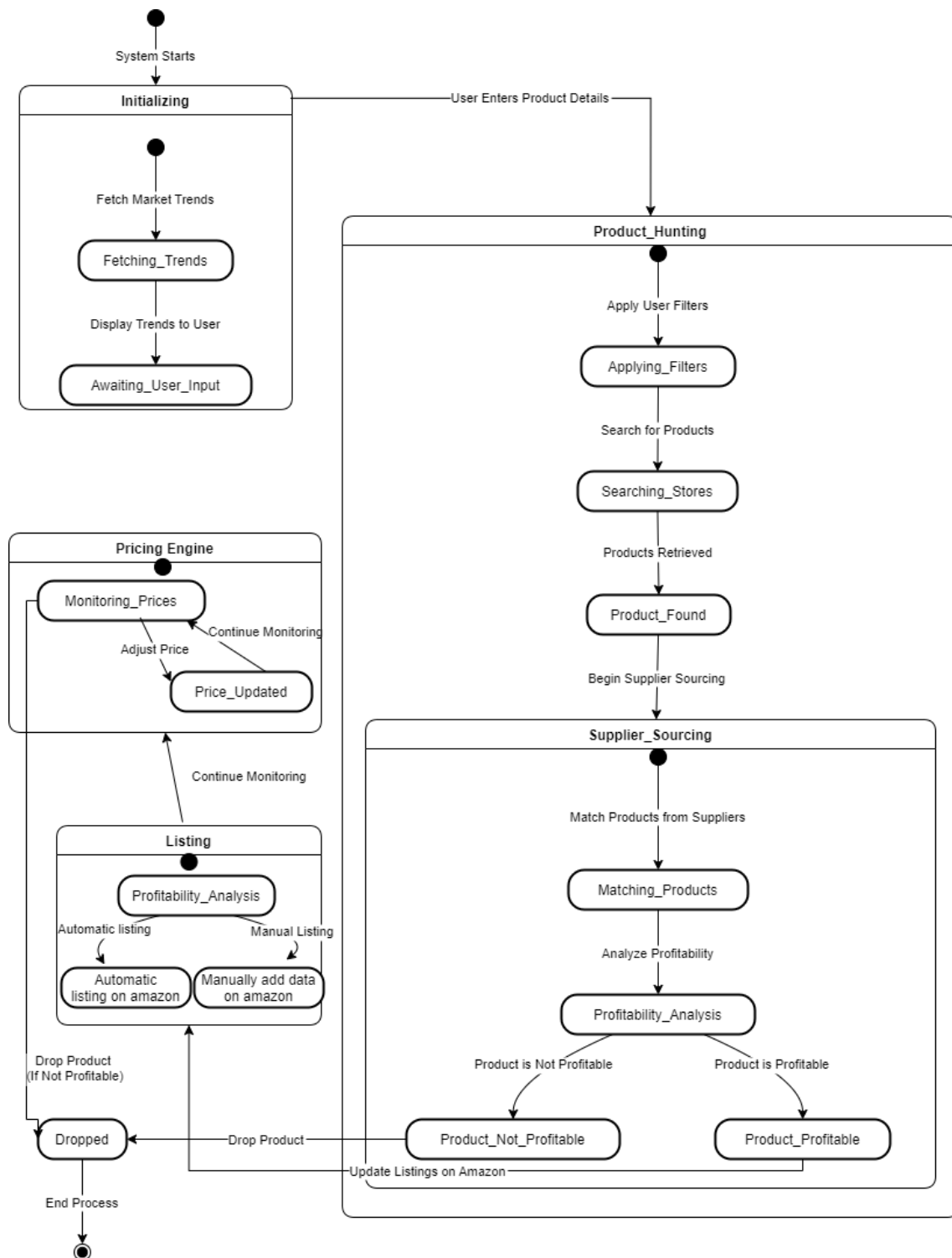


Figure 3.14: State Transition Diagram

3.3 Data Design

For efficient management of data we will be utilizing MongoDB. So given below are JSON Schema for our project.

```
"$schema": "http://json-schema.org/draft-07/schema#",
"title": "Amazon Seller Automation System",
"type": "object",
"properties": {
  "productInfo": {
    "title": "Product Information",
    "type": "object",
    "properties": {
      "productName": {
        "type": "string",
        "description": "The name of the product"
      },
      "productID": {
        "type": "string",
        "description": "The unique identifier for the product"
      },
      "category": {
        "type": "string",
        "description": "Category to which the product belongs"
      },
      "description": {
        "type": "string",
        "description": "Detailed description of the product"
      },
      "price": {
        "type": "number",
        "description": "Current price of the product"
      }
    }
  },
  "required": ["productName", "productID", "price"]
}
```

Figure 3.15: Product JSON schema

```
"supplierInfo": {
  "title": "Supplier Information",
  "type": "object",
  "properties": {
    "supplierName": {
      "type": "string",
      "description": "The name of the supplier"
    },
    "supplierID": {
      "type": "string",
      "description": "The unique identifier for the supplier"
    },
    "location": {
      "type": "string",
      "description": "Location of the supplier"
    },
    "availability": {
      "type": "boolean",
      "description": "Availability status of the product from the supplier"
    }
  },
  "required": ["supplierName", "supplierID", "availability"]
},
```

Figure 3.16: Supplier JSON schema

```
"profitabilityAnalysis": {  
  "title": "Profitability Analysis",  
  "type": "object",  
  "properties": {  
    "margin": {  
      "type": "number",  
      "description": "Profit margin of the product"  
    },  
    "salesTrend": {  
      "type": "string",  
      "description": "The recent sales trend of the product"  
    },  
    "competitorPrice": {  
      "type": "number",  
      "description": "Current competitor pricing for the product"  
    }  
  },  
  "required": ["margin"]  
},
```

Figure 3.17: ProfitAnalysis JSON schema


```
{
  "salesProfitTable": {
    "title": "Sales and Profit Table",
    "type": "object",
    "properties": {
      "salesData": {
        "type": "array",
        "items": {
          "type": "object",
          "properties": {
            "date": {
              "type": "string",
              "format": "date",
              "description": "Date of the sale"
            },
            "quantitySold": {
              "type": "number",
              "description": "Quantity of the product sold on the specified date"
            },
            "totalSales": {
              "type": "number",
              "description": "Total sales revenue for the specified date"
            },
            "profit": {
              "type": "number",
              "description": "Profit earned on the sales for the specified date"
            }
          },
          "required": ["date", "quantitySold", "totalSales", "profit"]
        }
      }
    },
    "required": ["salesData"]
  }
}
```

Figure 3.18: Sales Analytics JSON schema

```
"dynamicPricing": {  
  "title": "Dynamic Pricing",  
  "type": "object",  
  "properties": {  
    "priceUpdated": {  
      "type": "boolean",  
      "description": "Indicates if the price has been updated"  
    },  
    "newPrice": {  
      "type": "number",  
      "description": "The updated price after dynamic pricing adjustments"  
    }  
  },  
  "required": ["priceUpdated"]  
},
```

Figure 3.19: dynamicPrice JSON schema

Bibliography